3 An experiment is carried out, using a light intensity meter attached to a toy train, to measure the diffraction of laser light through a double slit, as shown in Fig. 3.1 below.

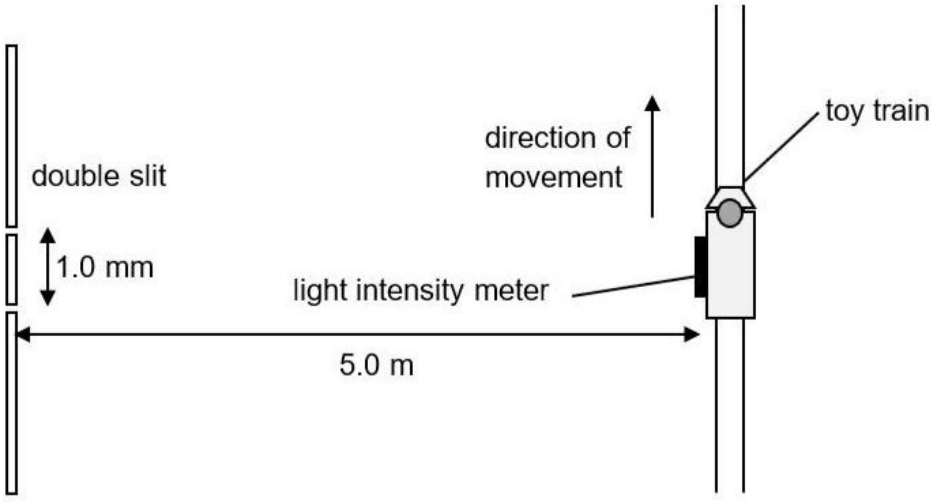


Fig. 3.1

Light from the laser is incident on the double slit. The light from the double slit is then detected by a light sensor attached to the train.

As the train moves along the track, the light intensity is recorded by the light intensity meter. The variation with time of the light intensity is shown in Fig. 3.2.

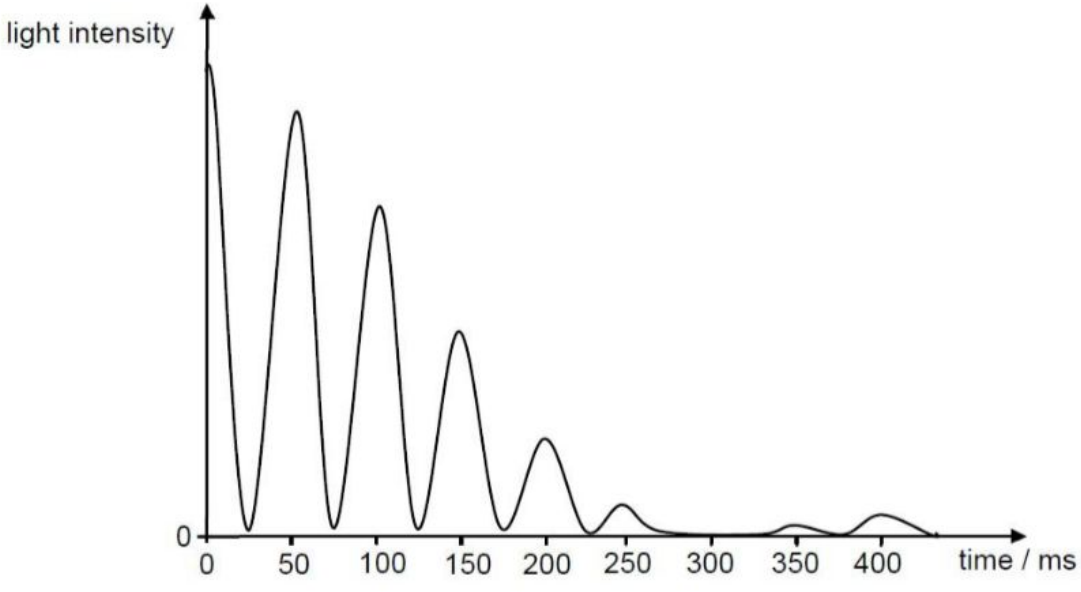


Fig. 3.2

(a) Explain, with reference to the paths travelled by the light emerging from each slit, why maxima and minima are observed.

Light from each slit travels a different distance to the sensor. When the path difference from each slit to the light intensity sensor is $n\lambda$ where n is an integer, the waves meet in phase and constructive interference occurs, light intensity reaches maximum value. [B1]

When the path difference from each slit to the light intensity sensor is $\left(n+\frac{1}{2}\right)\lambda$ where n is an integer, the waves meet antiphase and destructive interference occurs, light intensity reaches minimum value. [B1]

..... [2]

(b) **(i)** The speed of the train is 6.5 cm s^{-1}.
Use Fig. 3.2 to show that the fringe separation of the two slit interference pattern is 3.3 × 10^{-3} m.

fringe separation, $x = \left(\frac{200\times10^{-3}\text{ s}}{4}\right)\times\left(6.5\times10^{-2}\text{ m s}^{-1}\right) = 3.25\times10^{-3}\text{ m}$ [B1]

(do not accept answer based on 50 ms for one fringe spacing directly)

[Students should measure over several fringe spacings and divide by the number of intervals to reduce the percentage uncertainty in the fringe separation.]

[1]

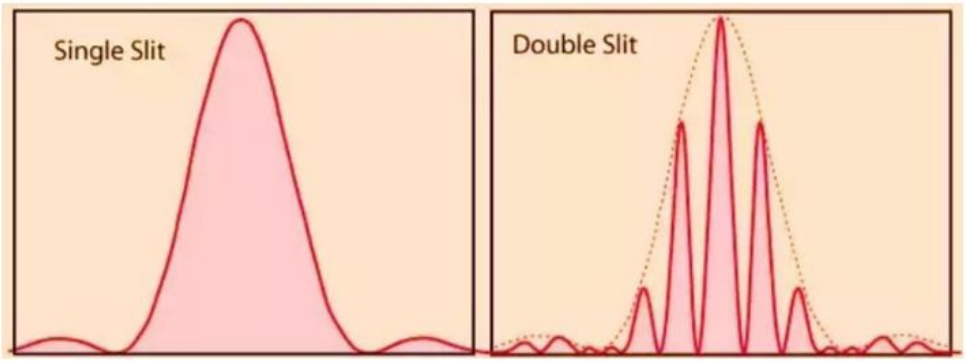
(ii) The separation of the slits is 1.0 mm and the track is 5.0 m from the slits.
Determine the wavelength of the laser light.

Since the slit-to-screen distance is much greater than the slit separation, i.e., $D \gg a$, we have $x = \frac{\lambda D}{a}$. Therefore,

$$\lambda = \frac{ax}{D} = \frac{\left(1.0\times10^{-3}\right)\left(3.25\times10^{-3}\right)}{5.0}$$
$$= 6.5\times10^{-7}\text{ m}$$

wavelength = ... m [2]

(c) A maximum is detected by the light intensity meter at every 50 ms, explain why a minimum is detected at 300 ms in Fig. 3.2.



The double slit interference pattern is modulated / modified / enveloped by the single slit diffraction pattern.

OR There is an envelope because of (single) slit diffraction

At 300 ms the light sensor is at the first minimum of the single slit diffraction envelope where the light intensity is zero (due to destructive interference between wavelets from different parts of each slit), so no maximum interference fringe is observed at that position. [B1]

..... [1]

(d) Determine the width of one of the slits.

position of first minimum, $y = (300 \times 10^{-3}\ \text{s})(6.5 \times 10^{-2}\ \text{m s}^{-1}) = 0.0195\ \text{m}$

single-slit diffraction first minimum: $b \sin\theta = \lambda$

small angle approximation: $\sin\theta \approx \tan\theta = \dfrac{y}{D}$

$\therefore b\left(\dfrac{y}{D}\right) \approx \lambda$

$\Rightarrow b \approx \dfrac{\lambda D}{y} = \dfrac{(650 \times 10^{-9})(5.0)}{0.0195} = 1.7 \times 10^{-4}\ \text{m}$

width = ... m [2]

(e) The experiment is changed so that the toy car is now used to investigate the formation of a stationary sound wave as shown in Fig. 3.3 below.

The light sensor of the train is replaced with a sound sensor attached to a data logger. A loudspeaker emitting sound waves of constant amplitude and frequency is placed such that it faces a reflecting barrier. The train track is adjusted so that the train is moving away from the loudspeaker towards the barrier.

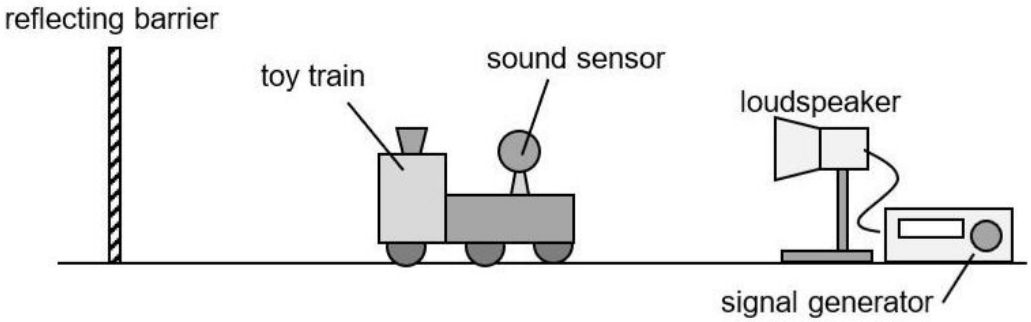


Fig. 3.3

The variation with distance of the sound intensity recorded by the sound sensor is shown in Fig. 3.4.

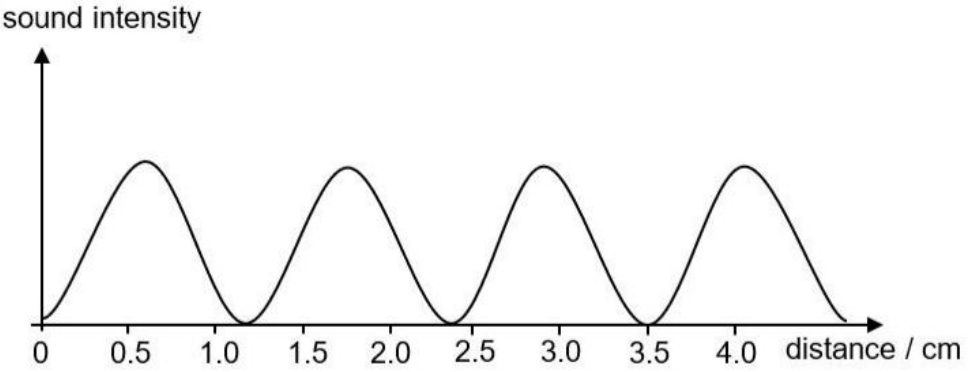


Fig. 3.4

(i) Explain how a stationary sound wave forms between the loudspeaker and reflecting barrier.

Sound wave from the loudspeaker is reflected by the barrier. [B1]

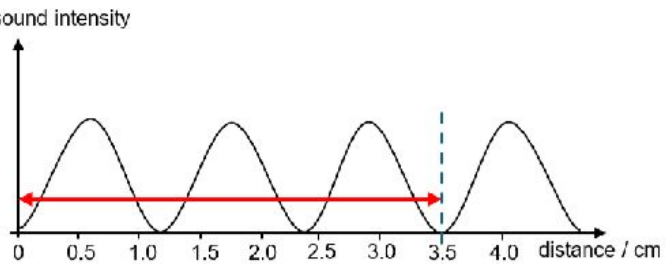
The reflected wave interferes / undergoes superposition with the incident wave. The resultant wave is a stationary wave. [B1]

..... [2]

(ii) Using Fig. 3.4, determine the wavelength of the stationary wave.

using separation between 4 consecutive nodes (1.5λ) as the distance from the graph, [B1]

wavelength, $\lambda = \left(3.5 \times 10^{-2}\ \text{m}\right) \div 1.5 = 0.0233\ \text{m}$ [A1]



wavelength = ... m [2]

(iii) Given that the frequency of the sound is 14 kHz.
Determine the speed of the sound wave.

speed, $v = f\lambda = (14000\ \text{Hz})(0.0233\ \text{m})$ [B1]

$= 326\ \text{m s}^{-1}$

$= 330\ \text{m s}^{-1}$ (to 2 sig. fig.) [A1]

speed = ... m s^{-1} [2]

[Total: 14]